ECOM90009 Quantitative Methods for Business

Workshop 4 (Week 5)

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Introduction

Zheng Fan

- Know more about myself: zhengfan.site
- Ph.D student in Economics at Unimelb

If you need any help, don't be shy

- Weekly consultation & Ed discussion board (your first priority)
- Email: `jnlye@unimelb.edu.au` for subject related issues
- Consult Stop 1 for special consideration
- Email: `fan.z@unimelb.edu.au`

Before asking any questions, make sure you have gone through the materials on Canvas and existing posts on Ed discussion board!

Assignment

Assignment 1

- worth up to 10% of your final grade for this subject.
- Submit by **Thursday, April 2, at 11.59 pm**.
- The word limit for this assignment is 1,000 words.
- Note that the "AI Usage Declaration" is not included in the word count.

No tutorial next week!

- Please enjoy and take a break
- Catch up materials

Introduction

- Continuous probability distributions
- Use standard normal distribution table (you need to be very comfortable with this)
- Sampling distributions for sample mean and proportion; and CLT

Question 1

Indicate whether each statement below is true or false. Justify your answer.

(a) In the normal distribution, the flatter the curve, the larger the standard deviation.

(b) In the normal distribution, the total area under the curve is equal to one.

(c) Using the standard normal curve, the probability or area between $z = -1.28$ and $z = 1.28$ is 0.1003.

(d) Let z_1 be a z-score that is unknown but identifiable by position and area. If the symmetrical area between $-z_1$ and $+z_1$ is 0.9544, the value of z_1 is 2.0.

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FALSE

Cumulative Standardized Normal Probabilities $P(-\infty < Z < z)$

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.00	0.5000	0.504	0.508	0.512	0.516	0.5199	0.5239	0.5279	0.5319	0.5359
0.10	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.20	0.5793	0.5832	0.5871	0.591	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.30	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.648	0.6517
0.40	0.6554	0.6591	0.6628	0.6664	0.67	0.6736	0.6772	0.6808	0.6844	0.6879
0.50	0.6915	0.695	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.719	0.7224
0.60	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.70	0.758	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.80	0.7881	0.791	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.90	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.834	0.8365	0.8389
1.00	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.10	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.877	0.879	0.881	0.883
1.20	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.9015
1.30	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177

Question 1(d)

Statement: *Let z_1 be a z-score that is unknown but identifiable by position and area. If the symmetrical area between $-z_1$ and $+z_1$ is 0.9544, the value of z_1 is 2.0.*

TRUE

Question 2

In a standard Normal distribution, what value(s) of z cut(s) off the region described? Find your answer by using the standard normal tables. Don't forget to draw a picture.

(a) The highest 20%

(b) The highest 75%

(c) The lowest 5%

(d) The middle 90%

Question 3

A cereal manufacturer has a machine that fills the boxes. Boxes are labelled "450 grams" so the company wants to have that much cereal in each box. But since no packaging process is perfect, there will always be minor variations. If the machine is set at exactly 450 grams and the Normal distribution applies, then about half of the boxes will be underweight making consumers unhappy and exposing the company to bad publicity. To prevent underweight boxes, the manufacturer has to set the mean a little higher than 450 grams. Based on their experiences with the packaging machine, the company believes that the amount of cereal in the boxes fits a Normal distribution with a standard deviation of 2 grams. The manufacturer decides to set the machine to put an average of 453 grams in each box. The company want no more than 4% of the boxes to be underweight.

Question 3

Let X be the weight of cereal in a box - $X \sim N(\mu = 453, \sigma = 2)$ - No more than 4% of the boxes to be underweight - Underweight if weight of cereal is below 450g

Use this information to answer the following questions.

(a) What fraction of the boxes will be underweight?

(b) Will the company's management be happy with the result you found in (a)? If not, what mean setting does the company need to use?

(c) If the company's management doesn't want to put any more cereal in each box, how can the machine be set to no higher than 453 grams and still have only 4% underweight boxes?

Question 4

Similar question, please practise yourself after the class.

Question 5

You run a shipping company that transports goods from China to the United States. The time it takes a shipment to reach the United States from China (in days) is normally distributed. The population of delivery times for all shipments has a mean of 40 days and a standard deviation of 6 days.

(a) What is the probability that one of your shipments is delivered more than 12 days later than average?

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- $P(X > 52) = P(Z > \quad)$

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Thus $\hat{p} \sim N(0.5, pq/n)$

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You can surely use what we learned before. There is also a short cut.

The 68-95-99.7 rule

In a unimodal, symmetric distribution about

- 68% of the values fall within 1 standard deviation of the mean,
- 95% fall within 2 standard deviations of the mean
- 99.7% - almost all - fall within 3 standard deviations of the mean.

Recall information from the question:

- $X \sim N(40, 6^2)$

Hence, there is approximately a probability of 0.95 that a randomly selected shipment will lie within two standard deviations ($12 = 6 \times 2$ days, $28 + 12 = 40$ and $52 - 12 = 40$) of the mean

So how many shipments would we expect to be delivered within 28 to 52 days?

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So how many shipments would we expect to be delivered within 28 to 52 days?

- We would expect approximately $10,000 \times 0.95 = 9,500$ of the shipments to be delivered within 28 to 52 days.

Question 5

- Let X be the time it takes a shipment to reach the United States from China (in days)
- $X \sim N(\mu = 40, \sigma = 6)$

(d) What is the probability that a random sample of 16 of your shipments has an **average delivery time** less than 37 days?

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$\bar{X} \sim N(\mu = 40, \sigma = 6/\sqrt{16} = 1.5)$

Thanks for your attention!

Let me know if there is any questions!